Linear and Quadratic Programming

Recognize the form, build the model, use a solver

Prof. Nipun Batra

Mathematical Foundations for AI · IIT Gandhinagar

Two common problem classes

A planning problem

A small workshop makes two products. One unit of product 1 earns 3, and one unit of product 2 earns 2.

Resource limits are

$$x_1 + x_2 \leq 4, \quad x_1 \leq 2, \quad x_2 \leq 3, \quad x_1, x_2 \geq 0.$$

The decision is

$$\max_{x_1, x_2} \quad 3x_1 + 2x_2.$$

Every expression is linear. This is a **linear program**.

Recognition matters more than hand calculation

Reliable solvers already exist for:

1. linear programs with millions of variables,
2. convex quadratic programs,
3. least-squares problems,
4. network flows, cone programs, and related structured classes.

Boyd and Vandenberghe call least squares, LP, and QP a *technology*: reliable, mature, ready to use. Our job is to decide:

1. what the variables mean,
2. what quantity should be optimized,
3. which statements must be constraints,
4. and whether the resulting class is convex.

Linear program

One common LP form is

$$\min_{\boldsymbol{x}} \quad \boldsymbol{c}^\top \boldsymbol{x}$$

$$\text{subject to} \quad A\boldsymbol{x} \leq \boldsymbol{b}, \quad E\boldsymbol{x} = \boldsymbol{d}.$$

The objective is linear. Every constraint is affine.

Variable bounds such as $\ell_i \leq x_i \leq u_i$ are also linear inequalities.

Quadratic program

A convex QP has the form

$$\min_{\boldsymbol{x}} \quad \frac{1}{2}\boldsymbol{x}^\top P\boldsymbol{x} + \boldsymbol{q}^\top \boldsymbol{x}$$

$$\text{subject to} \quad A\boldsymbol{x} \leq \boldsymbol{b}, \quad E\boldsymbol{x} = \boldsymbol{d},$$

where P is positive semidefinite ($P \geq 0$ in matrix order).

The feasible set is still described by linear constraints. The objective now has curvature.

Where they appear in AI and data science

problem	objective	class
diet / allocation	linear cost or reward	LP
transport / matching relaxation	linear cost	LP
least squares with bounds	quadratic error	QP
mean–variance portfolio	risk minus return	QP
hard-margin SVM	squared norm	QP

The support vector machine you will meet in ML is “just” the last row: its training problem is a convex QP, and the solver for it already exists.

Learning outcomes

By the end of this lecture you will be able to:

1. recognize an LP or convex QP from its objective and constraints,
2. model a two-food diet problem as an LP,
3. verify a small LP optimum by checking vertices,
4. derive and interpret the dual of that LP,
5. model projection and portfolio choice as QPs,
6. express both problem classes in CVXPY,
7. and read solver status, residuals, and dual values critically.

Linear objectives over polytopes

The feasible set is a polytope

For the workshop,

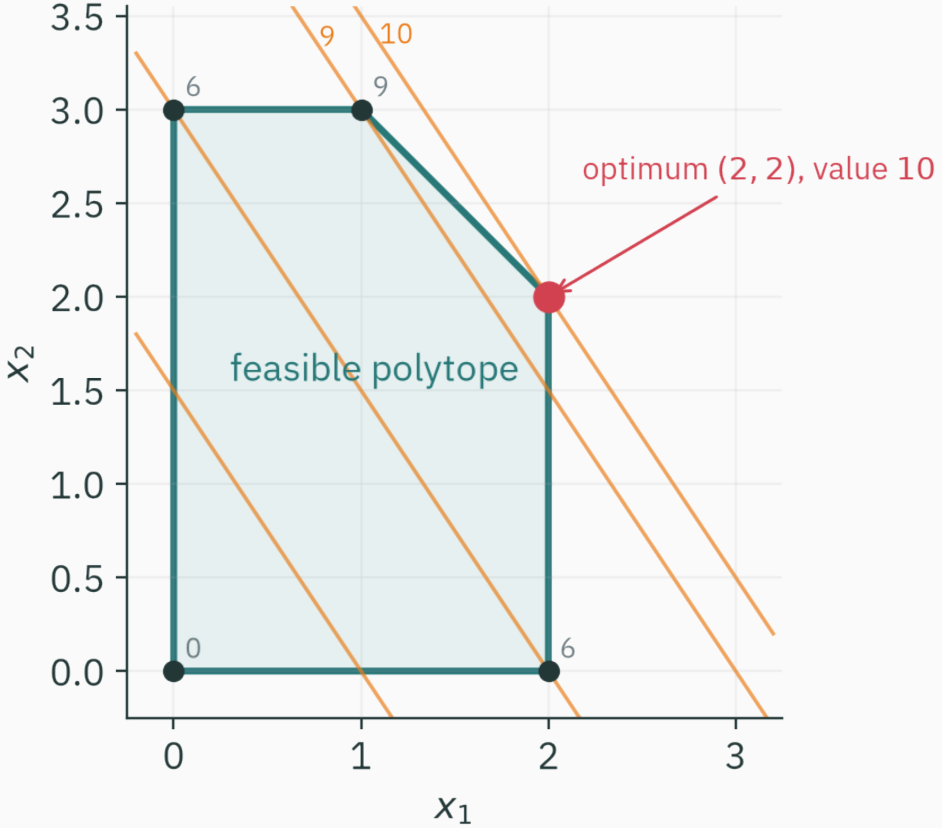
$$x_1 + x_2 \le 4, \quad x_1 \le 2, \quad x_2 \le 3, \quad x_1, x_2 \ge 0.$$

Each inequality defines a half-plane. Their intersection is a polygon in two dimensions, called a **polytope** in general dimensions.

Its vertices are

$$(0,0), (2,0), (2,2), (1,3), (0,3).$$

Move a linear contour



Contours of $3x_1 + 2x_2$ are parallel lines.

Moving the line toward larger values makes final contact at $(2, 2)$.

Check every vertex

D DERIVATION

vertex	$3x_1 + 2x_2$
$(0, 0)$	0
$(2, 0)$	6
$(2, 2)$	$\mathbf{10}$
$(1, 3)$	9
$(0, 3)$	6

For this bounded LP, a vertex attains the optimum.

Why a vertex is enough

D DERIVATION

Every point in a bounded polytope is a convex combination of its vertices:

$$\boldsymbol{x} = \sum_k \alpha_k \boldsymbol{v}_k, \quad \alpha_k \geq 0, \quad \sum_k \alpha_k = 1.$$

For a linear objective,

$$\boldsymbol{c}^\top \boldsymbol{x} = \sum_k \alpha_k \boldsymbol{c}^\top \boldsymbol{v}_k.$$

This weighted average cannot exceed the best vertex value. Hence some vertex is at least as good as $\boldsymbol{x}$.

Three possible LP outcomes

An LP may be:

1. **optimal:** a finite best value exists,
2. **infeasible:** no point satisfies all constraints,
3. **unbounded:** feasible points exist and the objective can improve without limit.

These are solver statuses, not numerical inconveniences.

Multiple optima

If an objective contour is parallel to a binding edge, every point on that edge can be optimal.

Example over the workshop polytope:

$$\max \quad x_1 + x_2.$$

The constraint $x_1 + x_2 \leq 4$ is tight along the segment from $(1, 3)$ to $(2, 2)$, so every point on that segment has value 4.

Question: where can an LP optimum occur?

Q QUESTION

A bounded feasible LP always has at least one optimum at...

A. the centre of the polytope

B. a vertex of the polytope

C. the origin

D. a point with $\nabla f = \mathbf{0}$

Answer

A ANSWER

Correct: B — a vertex of the polytope

Why: Every feasible point is a convex combination of vertices, so its linear-objective value cannot exceed the best vertex value.

From words to an LP

A two-food diet model

Choose servings of oats x and dal y.

	oats	**dal**
cost	2	3
protein	5	10
fibre	4	3

Requirements: at least 20 units of protein and 12 units of fibre.

Step 1 — choose variables and units

$$x = \text{servings of oats}, \quad y = \text{servings of dal}.$$

Write units beside the first draft:

1. cost: currency / serving,
2. protein: units / serving,
3. fibre: units / serving.

Then $5x + 10y$ has units of protein, as required.

Step 2 — write the objective

We want the least cost:

$$\min_{x,y} \quad 2x + 3y.$$

The coefficient of a variable is the change in objective per one unit of that variable.

If a statement cannot be assigned consistent units, the model is probably wrong.

Step 3 — translate requirements

"At least" becomes $\geq$:

$$5x + 10y \geq 20 \quad \text{(protein)},$$

$$4x + 3y \geq 12 \quad \text{(fibre)}.$$

Servings cannot be negative:

$$x \geq 0, \quad y \geq 0.$$

Solve the two-variable model by vertices

Candidate intersections on the lower boundary:

1. $y = 0$ gives $x = 4$, cost 8;
2. $x = 0$ gives $y = 4$, cost 12;
3. intersect $5x + 10y = 20$ — i.e. $x + 2y = 4$ — with $4x + 3y = 12$.

The intersection is

$$x = 2.4, \quad y = 0.8, \quad \text{cost} = 2(2.4) + 3(0.8) = 7.2.$$

The mixed diet is cheaper than either axis solution.

A modeling checklist

Before calling a solver:

1. State every variable in words and units.
2. Confirm the objective direction: minimize or maximize.
3. Translate "at most" as $\leq$ and "at least" as $\geq$.
4. Add physical bounds such as non-negativity.
5. Test one hand-constructed feasible point.
6. Estimate the expected scale of the answer.

A common modeling error

Suppose protein must be at least 20. Writing

$$5x + 10y \leq 20$$

describes a maximum allowance, not a minimum requirement.

The zero-food point would then be feasible and would minimize cost.

> If the solver returns a suspiciously perfect solution, inspect the model before blaming the solver.

Resource prices in the dual

The workshop LP in matrix form

Primal:

$$\max \quad \boldsymbol{c}^\top \boldsymbol{x} \quad \text{subject to} \quad A\boldsymbol{x} \le \boldsymbol{b}, \quad \boldsymbol{x} \ge \boldsymbol{0},$$

with

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}, \quad \boldsymbol{c} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}.$$

The three rows of A are the three resource limits.

The corresponding dual

For this maximization form, the dual is

$$\min_{\boldsymbol{\lambda}} \quad \boldsymbol{b}^\top \boldsymbol{\lambda}$$

$$\text{subject to} \quad A^\top \boldsymbol{\lambda} \geq \boldsymbol{c}, \quad \boldsymbol{\lambda} \geq \mathbf{0}.$$

Explicitly,

$$\min \quad 4\lambda_1 + 2\lambda_2 + 3\lambda_3$$

$$\text{subject to} \quad \lambda_1 + \lambda_2 \geq 3, \quad \lambda_1 + \lambda_3 \geq 2.$$

Read the dual variables as prices

$\lambda_1, \lambda_2, \lambda_3$ price the right-hand sides $4, 2, 3$.

The dual constraints require the total price of resources consumed by each product to cover its unit profit:

1. product 1 uses resources 1 and 2: $\lambda_1 + \lambda_2 \geq 3$,
2. product 2 uses resources 1 and 3: $\lambda_1 + \lambda_3 \geq 2$.

The dual minimizes the total priced capacity.

Solve the dual from complementary slackness

At the primal optimum $(x_1, x_2) = (2, 2)$:

1. constraints 1 and 2 are active,
2. constraint 3 has slack $3 - 2 = 1$.

Hence $\lambda_3 = 0$.

Both primal variables are positive, so both dual inequalities are tight:

$$\lambda_1 + \lambda_2 = 3, \quad \lambda_1 + \lambda_3 = 2.$$

Matching primal and dual values

From $\lambda_3 = 0$,

$\lambda_1 = 2, \quad \lambda_2 = 1.$

The dual value is

$4(2) + 2(1) + 3(0) = 10.$

The primal value at $(2, 2)$ is also

$3(2) + 2(2) = 10.$

Primal value = dual value = 10: a certificate of optimality.

Sensitivity from the dual

Near the current solution, one extra unit of resource 1 is worth about $\lambda_1 = 2$ objective units.

One extra unit of resource 2 is worth about $\lambda_2 = 1$.

Resource 3 has slack, so $\lambda_3 = 0$: a small increase has no value until another constraint becomes active.

This is L19′s shadow-price interpretation and L20′s inactive-multiplier rule in an LP.

A curved objective over the same polytope

Projection is a QP

Given a point $\boldsymbol{z}$, find its closest feasible point:

$$\min_{\boldsymbol{x}} \quad \frac{1}{2}\|\boldsymbol{x} - \boldsymbol{z}\|_2^2$$

$$\text{subject to} \quad A\boldsymbol{x} \leq \boldsymbol{b}.$$

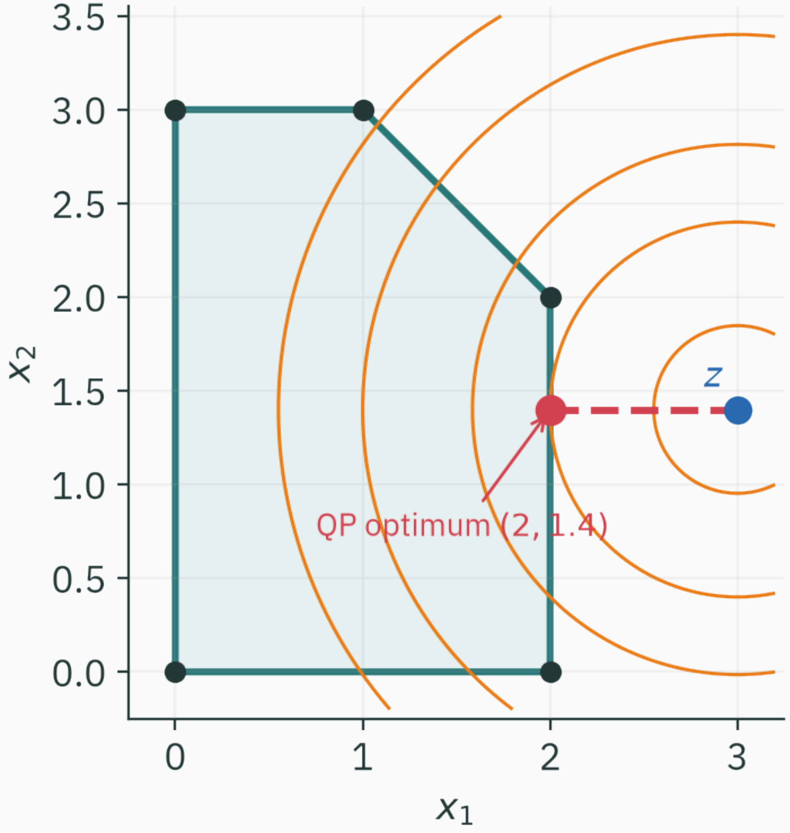
Expanding the objective:

$$\frac{1}{2}\boldsymbol{x}^\top \boldsymbol{x} - \boldsymbol{z}^\top \boldsymbol{x} + \frac{1}{2}\boldsymbol{z}^\top \boldsymbol{z}.$$

The final term is constant, so this is a QP with $P = I$ and $\boldsymbol{q} = -\boldsymbol{z}$.

A QP optimum need not be a vertex

V VISUAL



For $z = (3, 1.4)$, the projection is $(2, 1.4)$ — inside an edge, not at a vertex.

LP and QP geometry

	LP	**convex QP**
objective contours	parallel hyperplanes	ellipses / cylinders
curvature	none	P positive semidefinite
typical optimum	vertex, for a bounded polytope	interior, edge, or vertex
constraints	affine	affine

Why positive semidefinite P matters

The Hessian of

$$\frac{1}{2}\boldsymbol{x}^\top P\boldsymbol{x} + \boldsymbol{q}^\top \boldsymbol{x}$$

is P when P is symmetric.

If P is positive semidefinite, the objective is convex. Combined with affine constraints, every local optimum is global.

If P has a negative eigenvalue, the QP is non-convex and the same guarantees no longer apply.

Least squares with linear constraints

$$\min_{\boldsymbol{x}} \quad \frac{1}{2}\|A\boldsymbol{x} - \boldsymbol{b}\|_2^2 \quad \text{subject to} \quad C\boldsymbol{x} \leq \boldsymbol{d}.$$

Expand:

$$\frac{1}{2}\boldsymbol{x}^\top A^\top A\boldsymbol{x} - \boldsymbol{b}^\top A\boldsymbol{x} + \text{constant}.$$

Thus

$$P = A^\top A, \quad \boldsymbol{q} = -A^\top \boldsymbol{b}, \quad \text{and} \quad P \ \text{ is positive semidefinite.}$$

Least squares with affine equality or inequality constraints is a convex QP.

Balancing return and risk

Portfolio variables

Let w_i be the fraction invested in asset i.

$$\sum_i w_i = 1, \quad w_i \geq 0.$$

If expected returns are $\boldsymbol{\mu}$ and the covariance matrix is Σ:

1. expected portfolio return: $\boldsymbol{\mu}^\top \boldsymbol{w}$,
2. portfolio variance: $\boldsymbol{w}^\top \Sigma \boldsymbol{w}$.

Two common formulations

Minimum variance for a target return:

$$\min_{\boldsymbol{w}} \quad \boldsymbol{w}^\top \Sigma \boldsymbol{w}$$

$$\text{subject to} \quad \boldsymbol{\mu}^\top \boldsymbol{w} \geq r, \quad \mathbf{1}^\top \boldsymbol{w} = 1, \quad \boldsymbol{w} \geq \mathbf{0}.$$

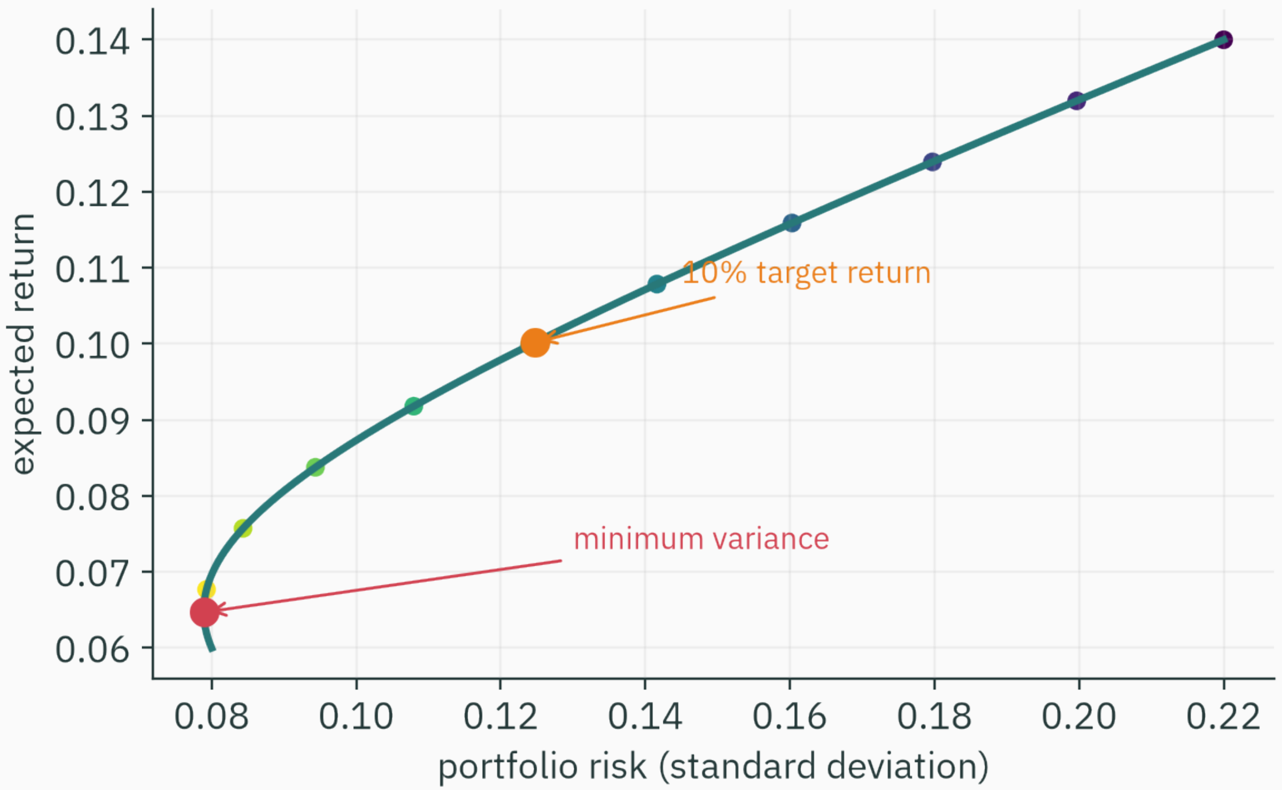
Or trade risk against return directly:

$$\min_{\boldsymbol{w}} \quad \frac{1}{2}\boldsymbol{w}^\top \Sigma \boldsymbol{w} - \gamma \boldsymbol{\mu}^\top \boldsymbol{w}.$$

Both are convex QPs when Σ is positive semidefinite.

The risk–return curve

V VISUAL



Changing the target return moves the solution along the feasible risk–return curve.

What the model omits

A useful mathematical model is still a simplification.

The basic mean–variance QP omits:

1. transaction costs,
2. uncertain estimates of $\boldsymbol{\mu}$ and Σ,
3. liquidity and position limits,
4. tail risk and non-Gaussian returns,
5. integer purchase sizes.

Some omissions remain convex after modeling; others change the problem class.

Express the model, then verify the result

Workshop LP in CVXPY

```python
import cvxpy as cp
import numpy as np

x = cp.Variable(2, nonneg=True)
c = np.array([3., 2.])
A = np.array([[1., 1.], [1., 0.], [0., 1.]])
b = np.array([4., 2., 3.])

problem = cp.Problem(cp.Maximize(c @ x), [A @ x <= b])
problem.solve()
print(x.value, problem.value)  # [2. 2.], 10.0
```

Read status before reading values

```
print(problem.status)
# 'optimal', 'infeasible', 'unbounded',
# or an '..._inaccurate' variant
```

Only interpret variable values after checking the status.

For an `optimal_inaccurate` result, inspect residuals, scaling, and solver tolerances.

Verify feasibility yourself

```
x_hat = x.value
print(A @ x_hat - b)   # every entry should be <= tolerance
print(x_hat.min())     # should be >= -tolerance
print(c @ x_hat)       # recompute the objective
```

The solver is a numerical component. Its output should satisfy the mathematical model to an appropriate tolerance.

Inspect dual values

```
constraint = (A @ x <= b)
problem = cp.Problem(cp.Maximize(c @ x), [constraint])
problem.solve()
print(constraint.dual_value)
# approximately [2., 1., 0.]
```

The last resource has slack and zero price, exactly as the hand calculation predicted.

Projection QP in CVXPY

```python
z = np.array([3.0, 1.4])
x = cp.Variable(2)

problem = cp.Problem(
    cp.Minimize(0.5 * cp.sum_squares(x-z)),
    [A @ x <= b, x >= 0],
)
problem.solve()
print(x.value)  # approximately [2.0, 1.4]
```

Use expressions that reveal structure

Prefer

```
cp.sum_squares(A @ x - b)
cp.quad_form(w, Sigma)
cp.norm(x, 2)
```

over manual algebra that obscures convexity.

CVXPY checks disciplined convex programming rules before choosing a solver.

Scaling affects numerical accuracy

If one coefficient is 10^9 and another is 10^{-7}, residuals and search directions become difficult to balance.

Useful steps:

1. choose consistent units,
2. scale variables to comparable magnitudes,
3. avoid redundant constraints,
4. use solver tolerances appropriate to the application,
5. and verify the unscaled answer afterward.

Question: classify the problem

Q QUESTION

Minimize $\|Ax - b\|_2^2$ subject to $x \geq 0$. What class is it?

A. LP

B. convex QP

C. non-convex QP always

D. unconstrained problem

Answer

A ANSWER

Correct: B — convex QP

Why: The squared least-squares objective has positive-semidefinite Hessian $2A^\top A$, and non-negativity is a linear constraint.

What the solver does

Choosing a solver

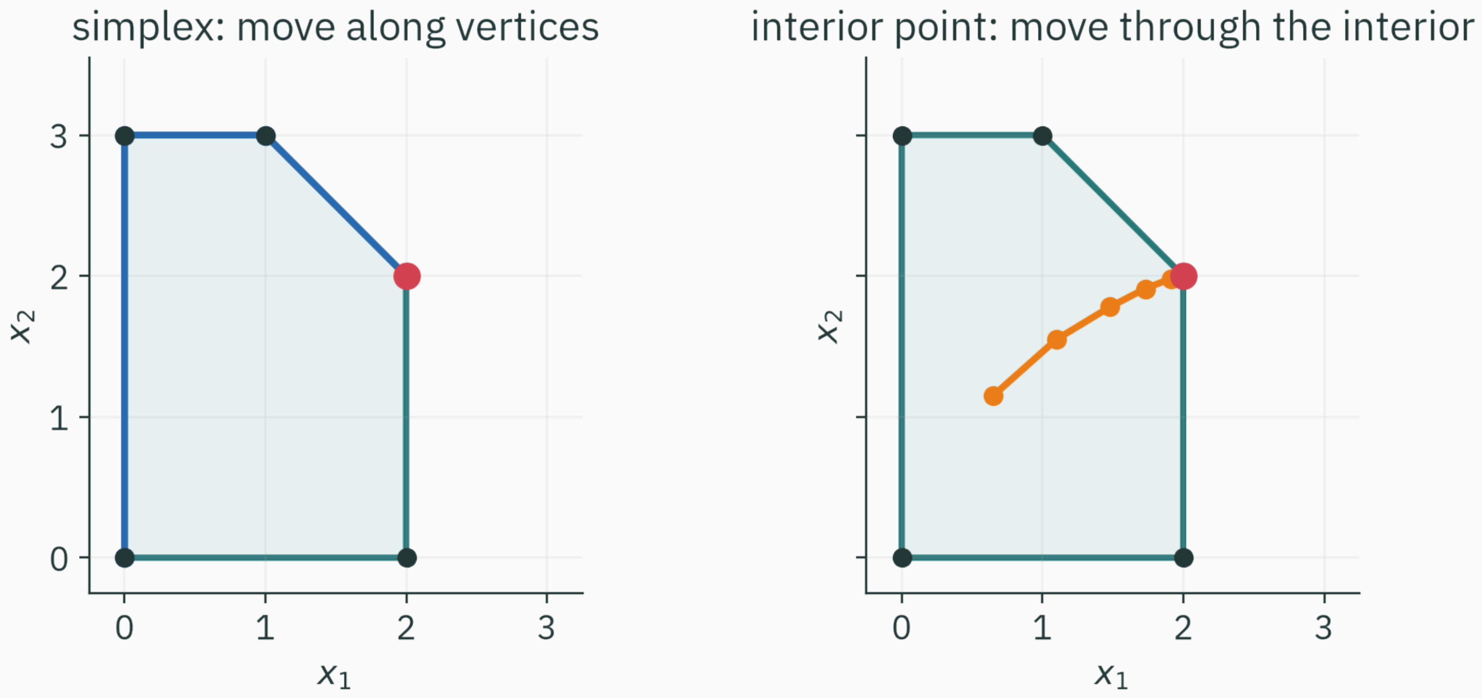
Choice depends on:

1. LP or QP structure,
2. dense versus sparse matrices,
3. problem size,
4. desired accuracy,
5. warm starts and repeated solves,
6. available commercial or open-source software.

Modeling systems can select a compatible solver; production work should record the solver and settings.

Two paths through an LP

★ OPTIONAL



The paths differ, but both exploit the LP's geometry and return the same optimum.

Simplex method, one-slide view

★ OPTIONAL

The simplex method:

1. starts at a feasible vertex,
2. chooses an adjacent edge that improves the objective,
3. moves to the next vertex,
4. stops when no adjacent improving edge remains.

Its worst-case complexity is exponential, but it is highly effective on many practical LPs.

Interior-point method, one-slide view

★ OPTIONAL

An interior-point method adds a barrier against constraint boundaries, for example

$$-\mu \sum_i \log(b_i - \boldsymbol{a}_i^\top \boldsymbol{x}).$$

It solves a sequence of smooth problems while reducing μ.

The iterates move through the interior and approach the optimal boundary as the barrier weakens.

Model, solve, check

A complete workflow

1. Define variables, units, and physical meaning.
2. Write the objective and constraints from the application.
3. Classify the problem: LP, convex QP, or something else.
4. Construct and test a small feasible instance.
5. Solve with an appropriate numerical solver.
6. Check status, feasibility, objective, and dual values.
7. Perform sensitivity checks before using the decision.

Summary

Linear and convex quadratic programs are standard numerical technologies; the main skill is building the right model.

1. LP: linear objective and affine constraints; over a bounded polytope, some vertex is optimal.
2. Convex QP: positive-semidefinite quadratic objective over affine constraints.
3. Dual variables attach marginal prices to constraints.
4. CVXPY expresses the model; the solver status and residuals still require inspection.
5. Scaling and units are part of correctness.

Practice

1. Formulate an LP that maximizes calories subject to cost and protein limits.
2. For the workshop LP, find all optima of $\max x_1 + x_2$.
3. Classify $\min \|\boldsymbol{x}\|_2^2$ subject to $A\boldsymbol{x} = \boldsymbol{b}$.
4. Add upper bounds $w_i \leq 0.4$ to the portfolio QP. Do they change convexity?

Practice answers

A ANSWER

1. Use food servings as non-negative variables; calories form a linear objective and cost/protein statements form linear constraints.
2. Every point on $x_1 + x_2 = 4$ between $(1, 3)$ and $(2, 2)$.
3. A convex equality-constrained QP with $P = 2I$.
4. No. Upper bounds are linear inequalities, so the problem remains a convex QP.

Module 4 in one view

lecture	question	tool
L16	is local information trustworthy?	convexity
L17	how do slopes move us?	gradient descent
L18	how does curvature change the step?	Newton / Gauss–Newton
L19	what if an equality must hold?	Lagrange multipliers
L20	what if a boundary may bind?	duality / KKT
L21	can a solver exploit the form?	LP / QP

Next: information

Optimization asks which feasible choice is best. The next module asks how uncertainty itself can be measured.

We begin with a quantity already seen in L19:

$$H(\boldsymbol{p}) = -\sum_i p_i \log p_i.$$

Next lecture: surprise, bits, and entropy.

Reading: Boyd & Vandenberghe, *Convex Optimization* §4.3–4.4 (LP and QP) · the CVXPY example library at cvxpy.org.

Don’t solve optimization problems — recognize them.